So, H.C.F. of 116,690,151 and 427,863,887 = 38896717.

$$\frac{116,690,151}{427,863,887} = \frac{116690151 \div 38896717}{427863887 \div 38896717} = \frac{3}{11}.$$

9. Since division by 0 is undefined, so 0 cannot be a factor of any natural number.

Hence, H.C.F. of 0 and 6 is undefined.

10. H.C.F. = Product of lowest powers of common factors $= 2^2 \times 3^2 \times 5$.

11. H.C.F. = Product of lowest powers of common factors $= 3 \times 5 \times 7 = 105$.

12. $4 \times 27 \times 3125 = 2^2 \times 3^3 \times 5^5$;

$8 \times 9 \times 25 \times 7 = 2^3 \times 3^2 \times 5^2 \times 7$;

$16 \times 81 \times 5 \times 11 \times 49 = 2^4 \times 3^4 \times 5 \times 7^2 \times 11$.

$\therefore$ H.C.F. $= 2^2 \times 3^2 \times 5 = 180$.

13. $36 = 2^2 \times 3^2$; $84 = 2^2 \times 3 \times 7$.

$\therefore$ H.C.F. $= 2^2 \times 3 = 12$.

14. Since all the numbers formed are even, 2 is a common factor.

Also, H.C.F. of two of the numbers i.e., 48 and 490, is 2.

So, the H.C.F. of all the numbers formed is 2.

15. $204 = 2^2 \times 3 \times 17$;

$1190 = 2 \times 5 \times 7 \times 17$; $1445 = 5 \times 17^2$.

$\therefore$ H.C.F. = 17.

16. H.C.F. of 18 and 25 is 1. So, they are co-primes.

17.
```
2923 ) 3239 ( 1
       2923
       ----
        316 ) 2923 ( 9
              2844
              ----
                79) 316 ( 4
                    316
                    ---
                     ×
```

$\therefore$ H.C.F. = 79.

18.
```
3444 ) 3556 ( 1
       3444
       ----
        112 ) 3444 ( 30
              3360
              ----
                84) 112 ( 1
                    84
                    --
                    28 ) 84 ( 3
                         84
                         --
                          ×
```

$\therefore$ H.C.F. = 28.

19. L.C.M. = Product of highest powers of prime factors $= 2^5 \times 3^4 \times 5^3 \times 7^2 \times 11$.

20.

2	24	–	36	–	40
2	12	–	18	–	20
2	6	–	9	–	10
3	3	–	9	–	5
	1	–	3	–	5

L.C.M. $= 2 \times 2 \times 2 \times 3 \times 3 \times 5 = 360$.

21.

2	22	–	54	–	108	–	135	–	198
3	11	–	27	–	54	–	135	–	99
3	11	–	9	–	18	–	45	–	33
3	11	–	3	–	6	–	15	–	11
11	11	–	1	–	2	–	5	–	11
	1	–	1	–	2	–	5	–	1

L.C.M. $= 2 \times 3 \times 3 \times 3 \times 11 \times 2 \times 5 = 5940$.

22. H.C.F. of 148 and 185 is 37. $\therefore$ L.C.M. $= \left(\frac{148 \times 185}{37}\right) = 740$.

23. H.C.F. of fractions $= \frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}}$.

H.C.F. of $\frac{a}{b}, \frac{c}{d}, \frac{e}{f} = \frac{\text{H.C.F. of } a, c, e}{\text{L.C.M. of } b, d, f}$.

24. Required H.C.F. $= \frac{\text{H.C.F. of } 2, 8, 64, 10}{\text{L.C.M. of } 3, 9, 81, 27} = \frac{2}{81}$.

25. Required H.C.F. $= \frac{\text{H.C.F. of } 9, 12, 18, 21}{\text{L.C.M. of } 10, 25, 35, 40} = \frac{3}{1400}$.

26. Required L.C.M. $= \frac{\text{L.C.M. of } 1, 5, 2, 4}{\text{H.C.F. of } 3, 6, 9, 27} = \frac{20}{3}$.

27. Required L.C.M. $= \frac{\text{L.C.M. of } 2, 3, 4, 9}{\text{H.C.F. of } 3, 5, 7, 13} = \frac{36}{1} = 36$.

28. Required L.C.M. $= \frac{\text{L.C.M. of } 3, 6, 9}{\text{H.C.F. of } 4, 7, 8} = \frac{18}{1} = 18$.

29. Given numbers with two decimal places are: 1.75, 5.60 and 7.00. Without decimal places, these numbers are: 175, 560 and 700, whose H.C.F. is 35.

$\therefore$ H.C.F. of given numbers = 0.35.

30. Given numbers are 1.08, 0.36 and 0.90. H.C.F. of 108, 36 and 90 is 18.

$\therefore$ H.C.F. of given numbers = 0.18.

31. Given numbers are 0.54, 1.80 and 7.20. H.C.F. of 54, 180 and 720 is 18.

$\therefore$ H.C.F. of given numbers = 0.18.

32. Given numbers are 3.00, 2.70 and 0.09. L.C.M. of 300, 270 and 9 is 2700.

$\therefore$ L.C.M. of given numbers = 27.00 = 27.

33. L.C.M. of A and B is B; L.C.M. of B and C is C $\Rightarrow$ L.C.M. of A, B and C is C.

34. $3240 = 2^3 \times 3^4 \times 5$; $3600 = 2^4 \times 3^2 \times 5^2$;

H.C.F. $= 36 = 2^2 \times 3^2$.

Since H.C.F. is the product of lowest powers of common factors, so the third number must have $(2^2 \times 3^2)$ as its factor.

Since L.C.M. is the product of highest powers of common prime factors, so the third number must have 3^5 and 7^2 as its factors.

$\therefore$ Third number $= 2^2 \times 3^5 \times 7^2$.

35. Let the required numbers be x, $2x$ and $3x$. Then, their H.C.F. $= x$. So, $x = 12$.

$\therefore$ The numbers are 12, 24 and 36.